THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 105	Complete graph on n points.
2	**(a)** 25	$\lvert 30h - 5.5m \rvert$, smaller of the two.
3	**(d)** 5	Repeat the largest-minus-smallest step.
4	**(b)** 180	Total minus the largest.
5	**(a)** 1001	Divide 1000 by the number and round up.
6	**(d)** 37	Try triangular numbers.
7	**(b)** 1170	$360n + 90$.
8	**(c)** 20	List the sequence.
9	**(c)** 4	Divide by 3 (the symbols are in the same proportion).
10	**(a)** 19	Total minus primes minus 1.
11	**(b)** 800	Subtract the first 10 odd numbers, whose sum is 100.
12	**(d)** 180	Net turn.
13	**(a)** 10	The second digit can be any of 10 digits.
14	**(c)** 84	Multiply.
15	**(c)** 24	LCM.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Complete graph idea: each pair plays once (1); $\frac{14\times13}{2} = 91$ (1).	1+1
17	Hour hand $= 125^\circ$, minute hand $= 60^\circ$ (1); Smaller angle $= 65^\circ$ (1).	1+1
18	Largest minus smallest each time (1); $1234 \to 3087 \to 8352 \to 6174$ (1).	1+1
19	Milk $= 150 - 62 - 48$ (1); 40 people (1).	1+1
20	LCM of 8, 12, 18 (1); $= 72$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$a + (n-1)d = 365$ gives $n - 1 = (365 - 14) \div 9$ (1); $n = 40$ (1); Term before it $= 356$ (1).	1+1+1
22	Hour hand at 55° (1); Minute hand at 300° (1); Smaller angle between them 115° (1).	1+1+1
23	Form $abba$ with $a = 1$, $b = 5$; Number 1551 $= 1551$ (1); $1551 \div 11 = 141$ exactly (1); Palindrome reads the same backwards (1).	1+1+1
24	Week 1 total $= 61$; Week 2 total $= 60$ (1); Larger: Week 1 (1); Difference $= 1$ (1).	1+1+1
	OR: LCM of 48, 72, 108 is 432 seconds (1); That is 7 minutes 12 seconds (1); Time: 8:07:12 am (1).	1+1+1
25	First 20 odd numbers sum to 400 (1); First 25 odd numbers sum to 625 (1); Required sum $= 225$ (1).	1+1+1
	OR: Minutes $= 120 \div 6 = 20$ (1); Time is 2:20 (1); Angle between hands $= 50^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Difference between consecutive figures $= 6$ (1); Figure 15: $a = 4$, so dots $= 88$ (1); For 118 dots: $n - 1 = (118 - 4) \div 6$ (1); $n = 20$ (1); Verification by substitution (1).	1+1+1+1+1
	OR: (a) Hour hand $= 35^\circ$ (1); (b) Minute hand $= 60^\circ$ (1); (c) Difference $= 25^\circ$, smaller angle $= 25^\circ$ (1); (d) acute angle (2).	1+1+1+2
27	Number 12321 (1); Digit sum 9 (1); Divisible by 9 (1); Smallest number to add: 0 (1); Check: new digit sum is a multiple of 9 (1).	1+1+1+1+1

	OR: A $= 36$, B $= 72$, C $= 108$ (1); Total $= 216$ (1); D $= (216) \div 3 = 72$ (1); Height of D $= 6.0$ cm (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 39	$20^2 - 19^2$ is the next odd number.
2	**(c)** 30	The hands are placed symmetrically about the 12-6 line, so the angle is the same.
3	**(b)** 165	Add the numbers.
4	**(b)** 21	Add and divide by the number of values.
5	**(b)** 24	HCF of three numbers.
6	**(a)** 11	Divide the total increase by the number of steps.
7	**(d)** 7	The hour hand turns 0.5° per minute.
8	**(b)** 24	Subtract.
9	**(a)** 37	Subtract.
10	**(b)** 5	Digit sum must be a multiple of 9.
11	**(a)** 88	Add the triangular number.
12	**(b)** 4	Each line gives one straight angle.
13	**(d)** 4	Find the palindrome.
14	**(a)** 16	Add the heights.
15	**(d)** 997	Subtract the remainder first.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$n \times n = 961$ so $n = 31$ (1); 31th odd number $= 2 \times 31 - 1 = 61$ (1).	1+1
17	Total $= 360 \times 2 + 90 = 810^\circ$ (1); Right angles $= 9.00$ (about 9) (1).	1+1
18	Apply the rule (1); Terms: 15, maximum: 52 (1).	1+1
19	First bar $= 180$; Second bar height $= 6$ cm (1); Value $= 120$ (1).	1+1
20	The numbers include a multiple of 3 (1); Hence the product is divisible by 3 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Build the row; Row: 2, 3, 5, 8, 13, 21, 34, 55 (1); 8th number $= 55$ (1); Sum $= 141$ (1).	1+1+1
22	Pens $= 36$ (1); Notebooks $= 72$ (1); Total $= 108$ (1).	1+1+1
23	Subtract the remainder: 72 and 96 (1); HCF of 72 and 96 is 24 (1); The number is 24 (it must exceed 4) (1).	1+1+1
24	Row 6 has $T_6 = 21$ cubes (1); Total in 6 rows uses the idea of adding triangular numbers (1); Sum $= \frac{6(7)(8)}{6} = 56$ (1).	1+1+1
	OR: Slices $= 2 \times 2 = 4$ (1); Angle $= 360 \div 4 = 90^\circ$ (1); Type: right (1).	1+1+1
25	Arrangements: $4 \times 3 \times 2 \times 1 = 24$ (1); Largest 9752, smallest 2579 (1); Difference 7173 (1).	1+1+1
	OR: Mean $= 58 \div 12 = 29/6$ (1); Above mean: 8 values (1); Frequencies: 1: 1; 2: 1; 3: 1; 4: 1; 5: 4; 6: 1; 7: 2; 8: 1 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) Side $=$ HCF $= 12$ m (1); (ii) Beds $= 5 \times 3 = 15$ (1); (iii) Perimeter of one bed $= 48$ m (1); Total fencing $= 720$ m (1); Cost $=$ ₹3600 (1).	1+1+1+1+1
	OR: 12th term of A $= 5 + 11 \times 4 = 49$ (1); 12th term of B $= 20 + 11 \times 2 = 42$ (1); Larger: A (1); Difference $= 7$ (2).	1+1+1+2
27	Net $= 180 - 45 + 90 = 225^\circ$ (1); Direction 225° from North (1); To face North again: turn clockwise 135° more (1); Type of the final turn: reflex (2).	1+1+1+2
	OR: Reverse-and-add repeatedly (1); Sums: 143, 484 (1); Steps $= 2$ (1); Palindrome 484 has 3 digits (odd) (2).	1+1+1+2

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 2025	Square of the triangular number.
2	**(c)** 72	$360^\circ \div 5$.
3	**(c)** 968	Reverse and add.
4	**(b)** 8	Integer division.
5	**(c)** 41	List the primes after 10.
6	**(b)** 41	Each triangular number adds the next counting number.
7	**(b)** 150	Multiply the angle between neighbours.
8	**(c)** 20	Continue the process.
9	**(c)** 7.5	Divide by the scale.
10	**(b)** 97	Check numbers just below.
11	**(b)** 153	Complete graph on n points.
12	**(c)** 105	$\lvert 30h - 5.5m \rvert$, smaller of the two.
13	**(b)** 7	Repeat the largest-minus-smallest step.
14	**(d)** 265	Total minus the largest.
15	**(a)** 6	$330 = 2 \times 3 \times 5 \times 11$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Sum of first 24 odd numbers $= 576$ and first 9 odd numbers $= 81$ (1); Required sum $= 495$ (1).	1+1
17	Each pair of rays makes one angle (1); $\frac{8\times7}{2} = 28$ (1).	1+1
18	Digit sum $= 12 \times 5 = 60$ (1); Remainder $= 6$ (same as digit sum divided by 9) (1).	1+1
19	Range $= 5 - 0 = 5$ (1); Mean $= 15 \div 6 = 5/2$ (1).	1+1
20	$84 = 2^2 \times 3 \times 7$ (1); Needed multiplier: 15 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	LCM $= 24$ (1); Required number $= LCM + 5 = 29$ (1); Check each remainder (1).	1+1+1
22	$a + (n-1)d = 331$ gives $n - 1 = (331 - 23) \div 7$ (1); $n = 45$ (1); Term before it $= 324$ (1).	1+1+1
23	Hour hand at 95° (1); Minute hand at 60° (1); Smaller angle between them 35° (1).	1+1+1
24	Form $abba$ with $a = 2$, $b = 8$; Number 2882 $= 2882$ (1); $2882 \div 11 = 262$ exactly (1); Palindrome reads the same backwards (1).	1+1+1
	OR: Week 1 total $= 64$; Week 2 total $= 62$ (1); Larger: Week 1 (1); Difference $= 2$ (1).	1+1+1
25	LCM of 30, 45, 60 is 180 seconds (1); That is 3 minutes 0 seconds (1); Time: 8:03:00 am (1).	1+1+1
	OR: First 25 odd numbers sum to 625 (1); First 31 odd numbers sum to 961 (1); Required sum $= 336$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Pink $= 100 - 70 = 30$ (1); Blue fraction $= \frac{25}{100}$ (2); Symbols: red: 4, blue: 5, green: 3, yellow: 2, pink: 6 (2).	1+2+2
	OR: LCM $= 24$ (1); Smallest 4-digit multiple: 1008 (1); Largest 4-digit multiple: 9984 (1); Verify divisibility (2).	1+1+1+2
27	Difference between consecutive figures $= 9$ (1); Figure 15: $a = 5$, so dots $= 131$ (1); For 176 dots: $n - 1 = (176 - 5) \div 9$ (1); $n = 20$ (1); Verification by substitution (1).	1+1+1+1+1
	OR: (a) Hour hand $= 85^\circ$ (1); (b) Minute hand $= 300^\circ$ (1); (c) Difference $= 215^\circ$, smaller angle $= 145^\circ$ (1); (d) obtuse angle (2).	1+1+1+2

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